



XBOX RGB Install Guide and Usage Instructions

Installation:

Installation of XBOX RGB is fairly straightforward and can be unique to each person's XBOX or use case. Below are some general guidelines. Please note that wiring harness colors may be nonstandard and you should review the port layouts to confirm.

XBOX-RGB ROL Notes:

If your installing an XBOX-RGB ROL on your V1 or V2 board ensure you are running the latest firmware (1.8.0 or greater). Connect 5V to pin 1 of the ESP32, connect GND to pin 2 of the ESP32, and connect CH7 to pin 13 or GPIO10 of the ESP32.

Required tools:

Multi-meter

Wire: AWG 28 or 30GA

Soldering Iron

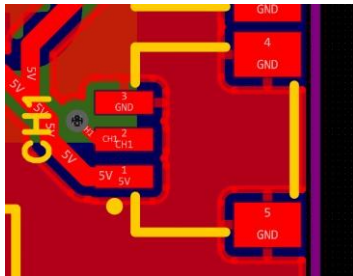
Double-sided tape

Heat shrink tubing (optional)

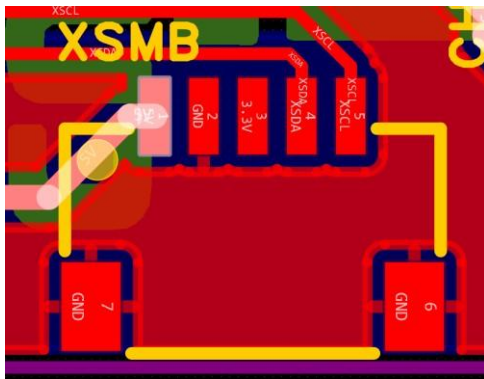
WS2812 LEDs (if you use your own LED strips)

Port Pinouts:

CH1 – 8:



XSMB:



V1 Notes:

V1 had a different connector layout from the production boards they will have CH1, CH2/3, CH4, CH5, and CH6.

V1 harness colors are as follows:

CH1, CH4, CH5 CH6:

Green: 5V

Yellow: RGB

Brown: GND

CH2/3:

Green: 5V

Yellow: RGB CH2

Blue: RGB CH3

Brown: GND

XSMB:

Red: 5V

Black: GND

Blue: SDA

White: SCL

V2 Notes:

V2 is the production board and the pinouts are as follows:

CH1, CH2, CH3, CH4, CH5, CH6

Yellow: 5V

Black: RGB

Red: GND

XSMB:

Blue: 5V

Green: GND

White: 3.3V (Not Used)

Black: SDA

Red: SCL

V3 Notes:

V3 is the current board with some features, here are the new additions:

CH1, CH2, CH3, CH4, CH5, CH6, CH7, CH8

Yellow: 5V

Black: RGB

Red: GND

XSMB:

Blue: 5V

Green: GND

White: 3.3V (Not Used)

Black: SDA

Red: SCL

Kratos_In is a test pad for future expansion

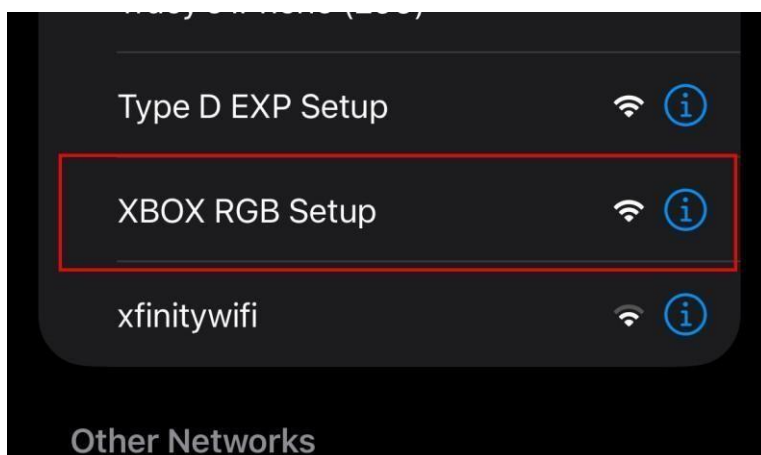
XBOX RGB only needs 5V and GND to function, and these can be tapped from a few sources. Use your multimeter to probe for a viable 5V source. SDA and SCL are required for the proper function of CH5 and CH6 if you choose to install those. SDA can be tapped from PIN14 and SCL from PIN13 of the VBOX LPC. You could even grab 5V and GND from there as well. For XBOX 1.6/b users DO NOT tap 5V from the LPC as your XBOX RGB kit will always be on. Below are examples of LCP pinouts and an ALT 5V point for 1.6/b XBOX's



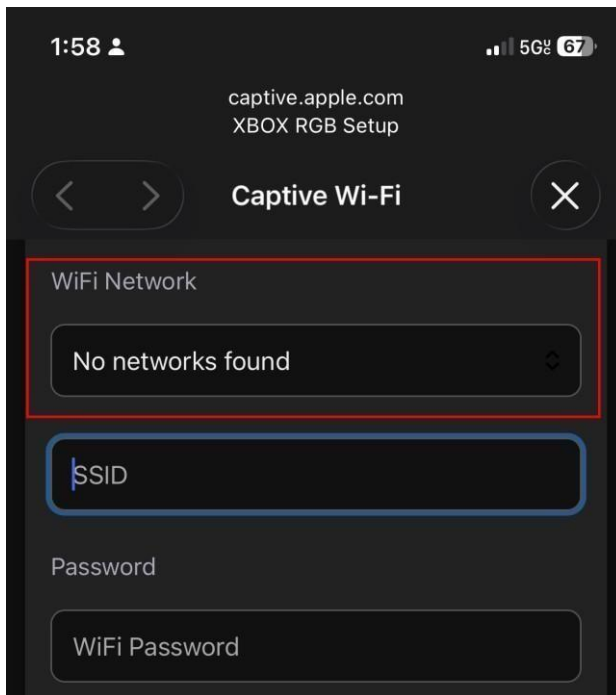
Connecting to WiFi:

After installing your XBOX RGB kit and on its first power up it will broadcast a WiFi network called XBOX RGB Setup. You can do a few things here to don't ever have to join it to your network if you don't want to, as you can tap on Open Config to hop right into the config menu and use it standalone.

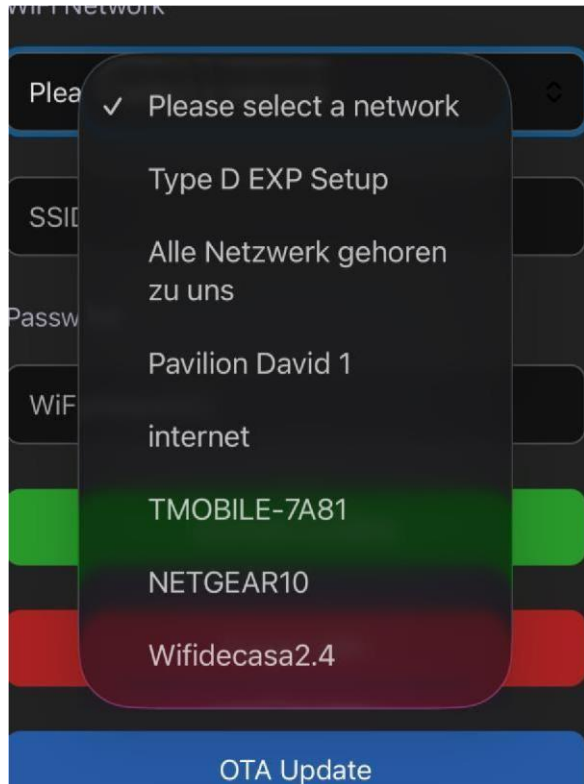
1. Tap on XBOX RGB Setup



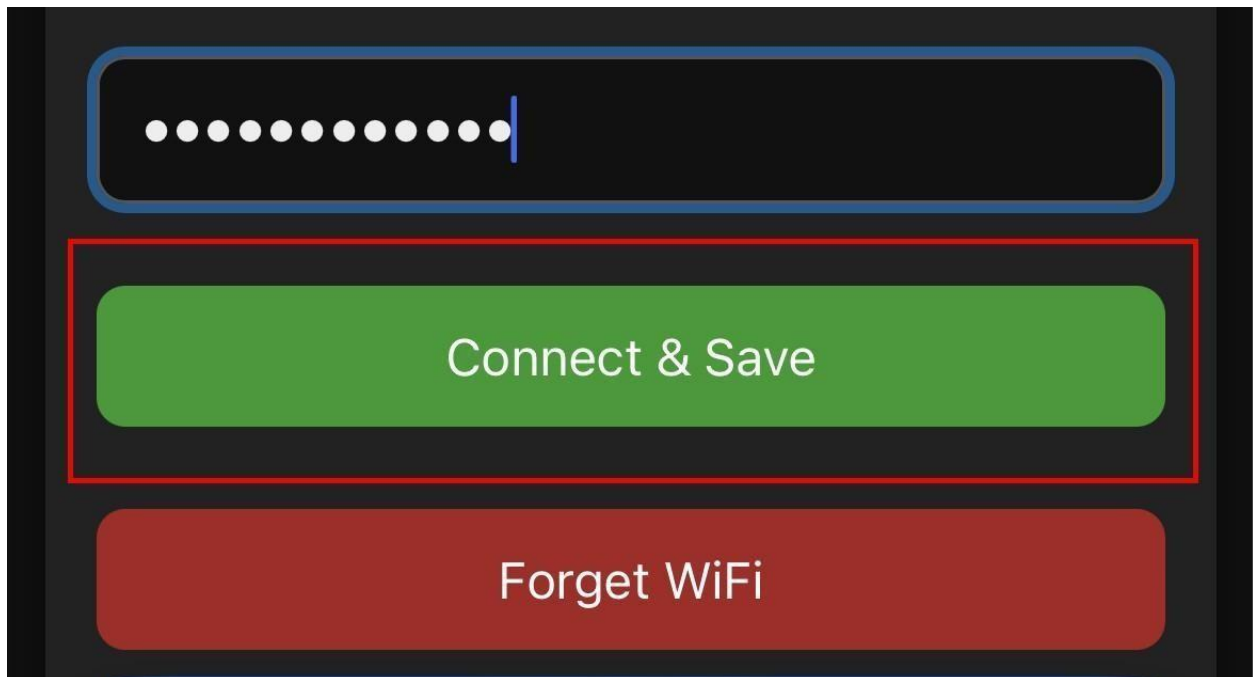
2. Tap on the No networks found section of WiFi networks



3. Select your WiFi network

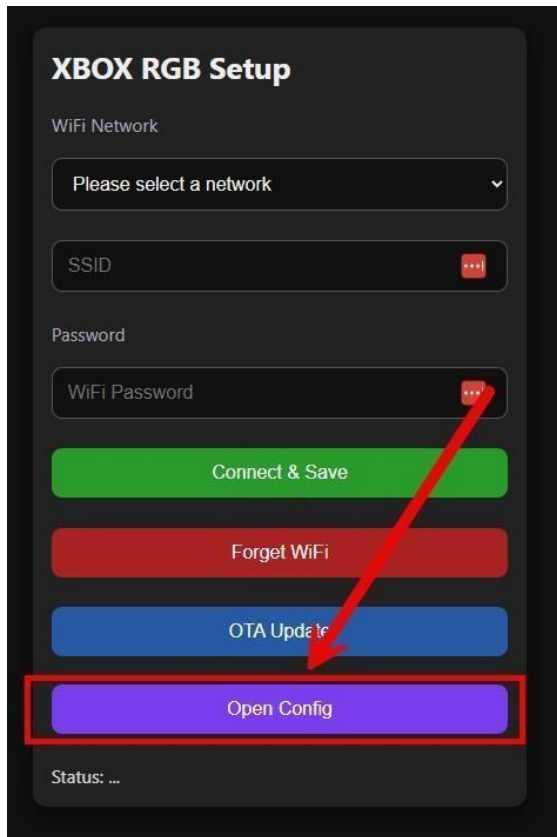


4. After selecting your WiFi network, tap on Connect & Save



Configuring your device:

To configure XBOX RGB, simply open a web browser and visit [HTTP://xboxrgb.local](http://xboxrgb.local). Once open tap on Open Config.



The image shows a mobile application interface titled "XBOX RGB Setup". It features a dark background with white text. The "WiFi Network" section includes a dropdown menu labeled "Please select a network" and a text input field for "SSID". The "Password" section includes a text input field for "WiFi Password". Below these fields are four buttons: "Connect & Save" (green), "Forget WiFi" (red), "OTA Update" (blue), and "Open Config" (purple). The "Open Config" button is highlighted with a red rectangular border, and a red arrow points to it from the "WiFi Password" field. At the bottom, there is a "Status: ..." label.

XBOX RGB Setup

WiFi Network

Please select a network

SSID

Password

WiFi Password

Connect & Save

Forget WiFi

OTA Update

Open Config

Status: ...

Configuration Overview:

RGB Controller (CH1-CH4) ready

Mode

Plasma

Brightness

Speed

Primary Color

Secondary Color

CH1 (Front) Count

8

CH2 (Left) Count

12

CH3 (Rear) Count

12

CH4 (Right) Count

12

Channel Direction

☒ Reverse CH1 (Front)

☐ Reverse CH2 (Left)

☐ Reverse CH3 (Rear)

☒ Reverse CH4 (Right)

Master

☐ Master Off (blank all channels)

Resume last mode on boot

Yes

Xbox SMBus LEDs

☐

Enable CPU temp LEDs (CH5)

☐

Enable Fan speed LEDs (CH6)

Disable to avoid SMBus polling by the other module.

Save

Reload

Reset Defaults

All changes preview live. Click Save to persist to flash.

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On the main configuration page, you have full control of XBOX RGB. Some key things to remember: you will preview modes instantly if you like a mode tap on save to save the settings. XBOX RGB will reload those settings on boot. You have to define your LEDs per channel. CH 1 – 4 can have a max of 50 LEDs per channel. If you install a strip backwards, you can reverse the direction of an individual channel in the config, Larson or Clock Spin are good animations to ensure you have good direction. Animations flow from left to right so CH 1 to 2 then to 3 etc.

Animation modes:

Solid

Breathe (synced across all channels)

Color Wipe

Larson Scanner (Cylon)

Rainbow (synced progression around the ring)

Theater Chase

Twinkle

Comet

Meteor (Color A→Color B trail)

Clock Spin

Plasma

Fire / Flicker

Palette Cycle

Palette Chase

Custom (Playlist) [You can string together animations in a playlist style]

USNC <-> Covenant

An additional feature of XBOX RGB is CH5 (CPU Temp) and CH6 (Fan Speed). These are optional, and to use them, you will need to wire SDA and SCL on the XSMB harness. If you have a Type D Expansion unit installed, these features will not work.

Additional ways to configure:

XBOX RGB also has a companion PC app along with an XBMC4Gamers script if you choose to use those. Those are available for free download on our github: <https://github.com/Darkone83/XBOX-RGB>.

Features:

- ✨ **12 animation modes** (see below) with live preview
- ☀️ Global **brightness** (1–255)
- 💡 **Power on fade in** slow but gentle boot fade in
- ⚙️ Per-mode controls: **Speed, Intensity, Width/Gap, Color A / Color B**
- 🔄 **Sane orientation & sync:** effects start at **CH1 (Front)** and flow clockwise **CH1→CH2→CH3→CH4**; all “phase” effects are synchronized
- 🔄 **Adjustable Strip length** CH1 - 4 can support up to 50 LEDs configurable from the webui
- 🔄 **Reversible channel support** Adjust the direction of your channels if you installed them backwards
- 💾 **Persistent settings** in NVS (counts, mode, colors, brightness, toggles...)
- 📶 **Wi-Fi portal** (AP SSID XBOX RGB Setup) with a sticky SSID picker
- 🌐 **Web UI** at /config
- 📄 **OTA updates** at /ota
- 📶 **SMBus (Xbox SMC) telemetry** (optional, on CH5/CH6) ○ **CPU temperature** bar (green→yellow→red, max 75 °C) ○ **Fan percentage** bar (blue→yellow→orange)
- 📶 **UDP control** UDP communication and control for XBMC4Gamers and the PC app